That's actually the smarter approach.

Right now, you're thinking in **days**, but admissions committees and researchers think in **milestones**.

Instead of Day 1–30, let's run this as a **12-week research project**.

🎯 Goal

By **September**, have:

✅ GitHub project

✅ Research paper draft

✅ Volatility forecasting models

✅ Visualizations

✅ Results section

✅ Strong portfolio project for MSc applications

Week 1 — Data & Research Foundation (Current Week)

Objective

Build a clean research dataset and establish benchmark performance.

Deliverables

Data

- Market data collection
- AAPL
- MSFT
- SPY
- VIX

Feature Engineering

Current feature set:

Target_GK_Vol

Target_Vol_t1

AAPL_Return

Ret_Lag1

Ret_Lag5

Ret_Lag10

Vol_Roll5

Vol_Roll20

Vol_Roll60

Vol_Change

VIX_Close

VIX_Change

SPY_Return

SPY_Vol

MSFT_Return

MSFT_Vol

AAPL_Volume

Volume_Change

EDA

- Correlation matrix
- Return distributions
- Volatility clustering
- VIX relationships

Baselines

- Naive Forecast

- Historical Average

Paper

Complete:

- Abstract
- Introduction
- Data
- Methodology
- Baseline Models

Week 2 — Econometric Models

Objective

Build traditional volatility models.

Models

GARCH(1,1)

Forecast next-day volatility.

EGARCH

Capture leverage effects.

Research question:

Does negative news increase volatility more than positive news?

Deliverables

- GARCH forecasts
- EGARCH forecasts
- QLIKE
- RMSE
- MAE

Paper

Write:

Section 5:

Econometric Models

Week 3 — Random Forest

Objective

First machine learning benchmark.

Models:

- Random Forest Regressor

Tasks:

- Hyperparameter tuning
- Feature importance
- Forecast evaluation

Deliverables

- Forecasts
- Feature importance chart

Paper

Add:

Machine Learning Models

(Random Forest)

Week 4 — XGBoost

Objective

Main machine learning model.

This will likely become your strongest model.

Tasks:

- Train XGBoost
- Hyperparameter optimization
- Compare against GARCH

Deliverables:

- Forecasts
- Importance plots
- Results tables

Week 5 — Model Comparison

Create final comparison table.

Example:

Model	QLIKE	RMSE	MAE
Naive			
Historical Avg			
GARCH			
EGARCH			
Random Forest			
XGBoost			

Week 6 — Realized Volatility Extension

Add Dataset 2.

Potential:

- HAR-RV
- Realized Volatility
- Oxford-Man style data

This is where the project starts looking graduate-level.

Week 7–8 — Research Paper

Complete:

- Literature Review
- Results

- Discussion
- Conclusion

Target:

10–15 pages.

Week 9–10 — GitHub Upgrade

Add:

- Visuals
- README
- Research summary
- Project architecture

Turn repository into a portfolio piece.

Week 11–12 — Applications

Use project for:

- MSc Financial Engineering
- MSc Finance
- MSc AI
- Quantitative Finance programs

This Week's Only Goal

Don't touch GARCH yet.

Finish Week 1 completely:

1. Run EDA
2. Run baseline models
3. Save figures
4. Save results
5. Update paper sections

Once you show me:

baseline_results

and your target correlations, we'll close Week 1 and start Week 2 with GARCH and EGARCH. That's the point where the project starts looking like something that can genuinely strengthen a top master's application. 🔥 📈